

Confirmatory Factor Models

Specification

Agenda

- Specifying the “typical” CFA
- Accounting for local dependence
- Accounting for multidimensionality
- 3 types of special CFAs

Why Do We Need CFAs?

- Our observed variables do not capture the relationship among our latent variable very well.
- CFAs allow us to estimate the relationships among the latent variables and observed variables
- CFAs provide the building blocks to full structural equation models which combine path analysis (SEM with observed variables) and CFA together to estimate the relationship among latent variables
- CFA is also useful in its own right for measure development and evaluation
 - Not the focus of this class

Steps of CFA

- 1. Model specification**

Translate your hypothesis into a model

- 2. Identification**

Make sure your model is mathematically estimable

- 3. Parameter Estimation**

Fit your model with software

- 4. Model Evaluation**

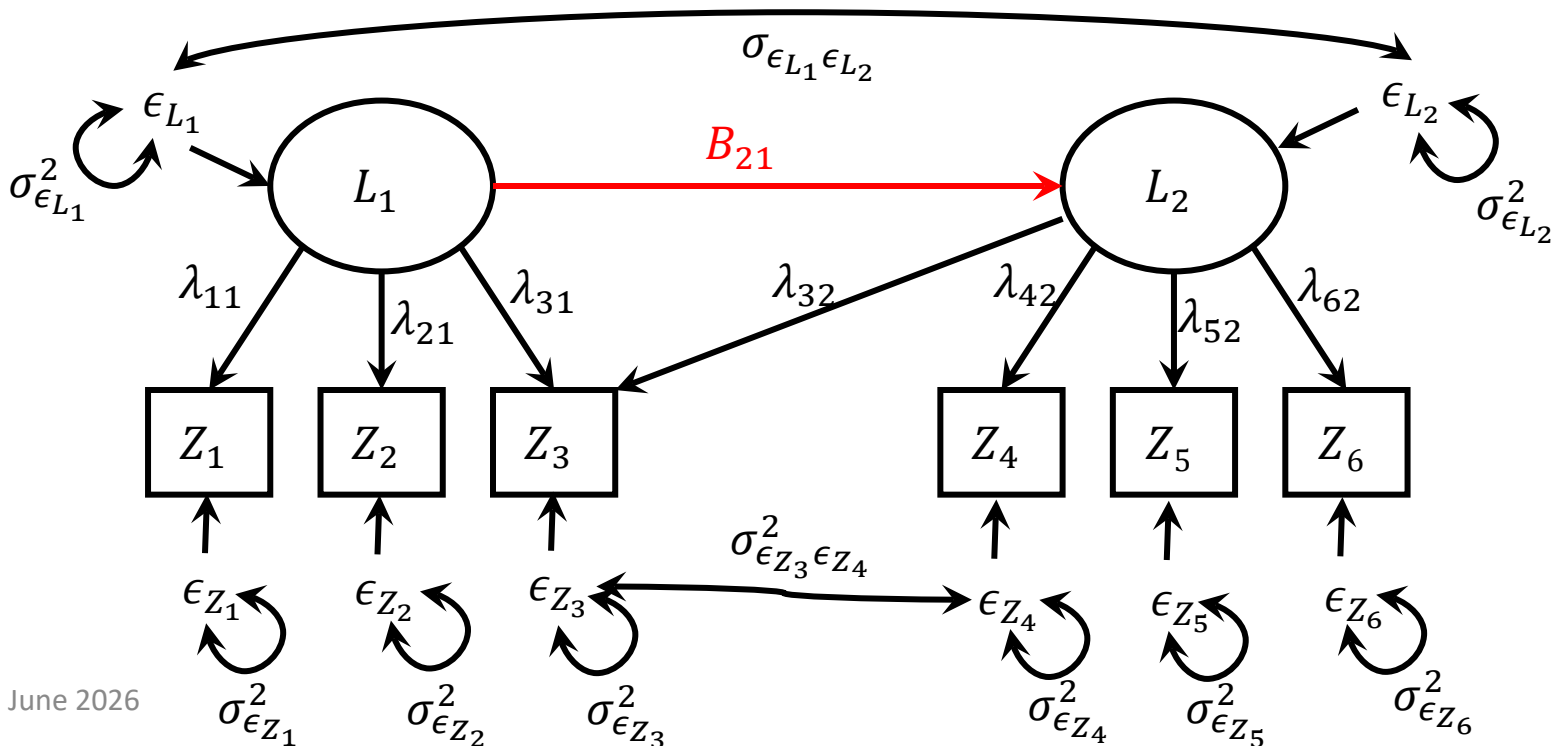
Assess how well your model fits the observed sample data/covariance

- 5. Model Modification**

Evaluate potential modifications to the model

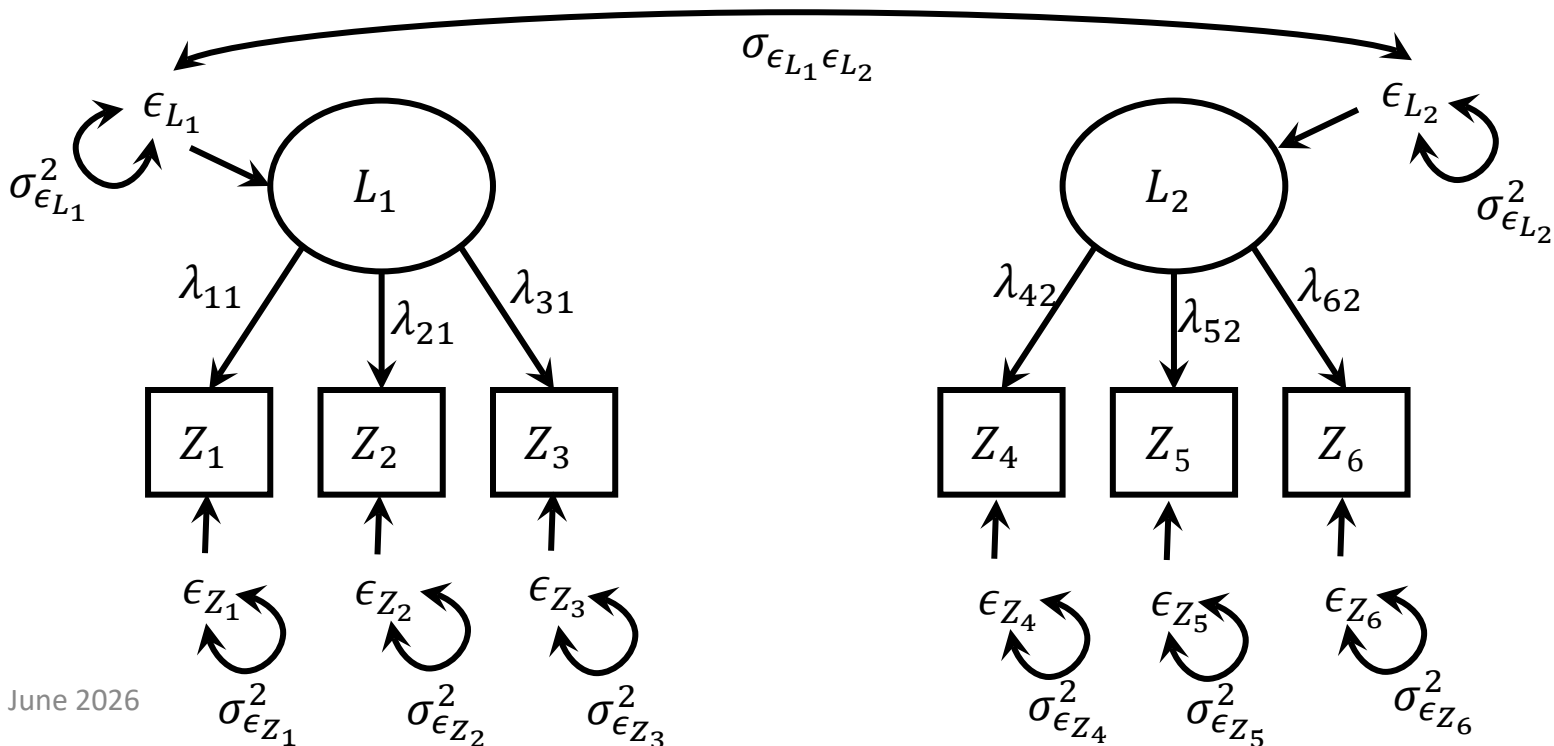
What is a Confirmatory Factor Analysis (CFA)?

- A confirmatory factor analysis is a model where observed variables are caused by a latent (unobserved) factor.
- There can be multiple observed variables and multiple factors
- Factors can be correlated with each other
 - but cannot have directed paths
- Observed variables can be caused by multiple factors (cross-loading)
- Deviances can have “residual correlations”



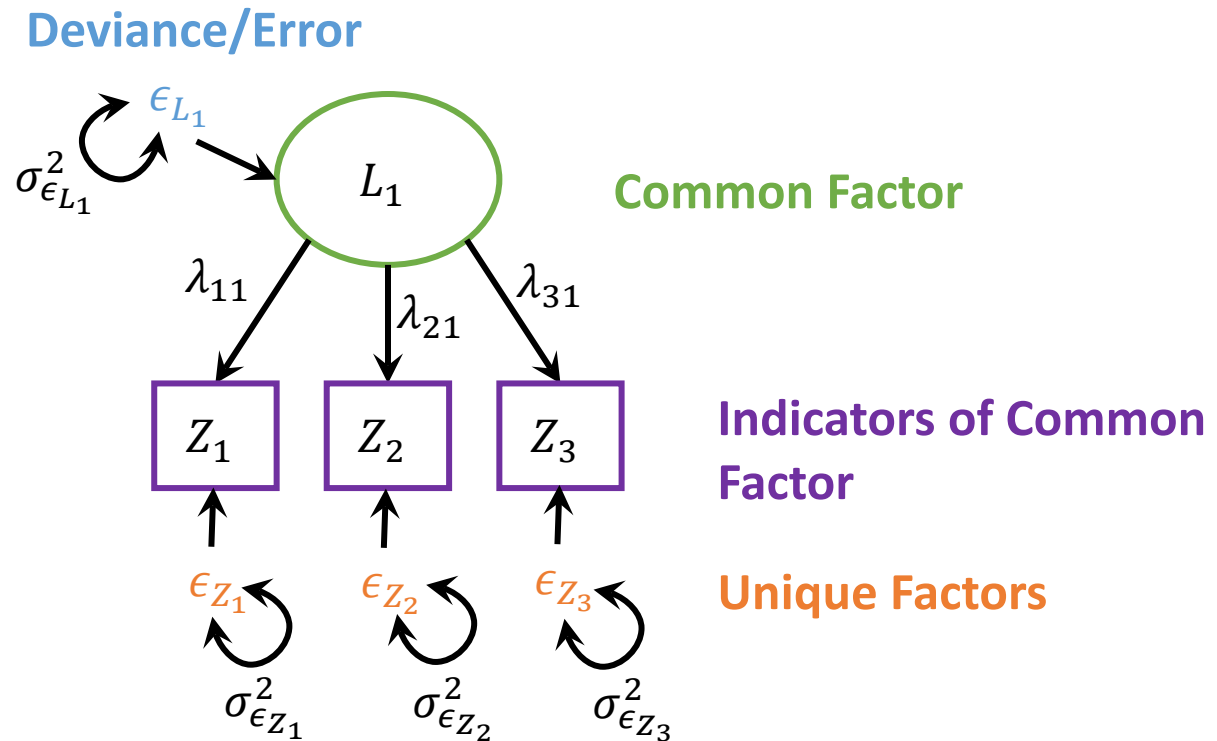
A note on notation

- All latent variables are **exogenous**
- Observed variables are Z (they still need deviations ϵ_Z)
- Variances and covariances among latent variables are always σ_{ϵ_L}
- Variances and covariances of deviances are just σ_{ϵ_Z}
- To-from notation for the λ s



Common vs. Unique Factors

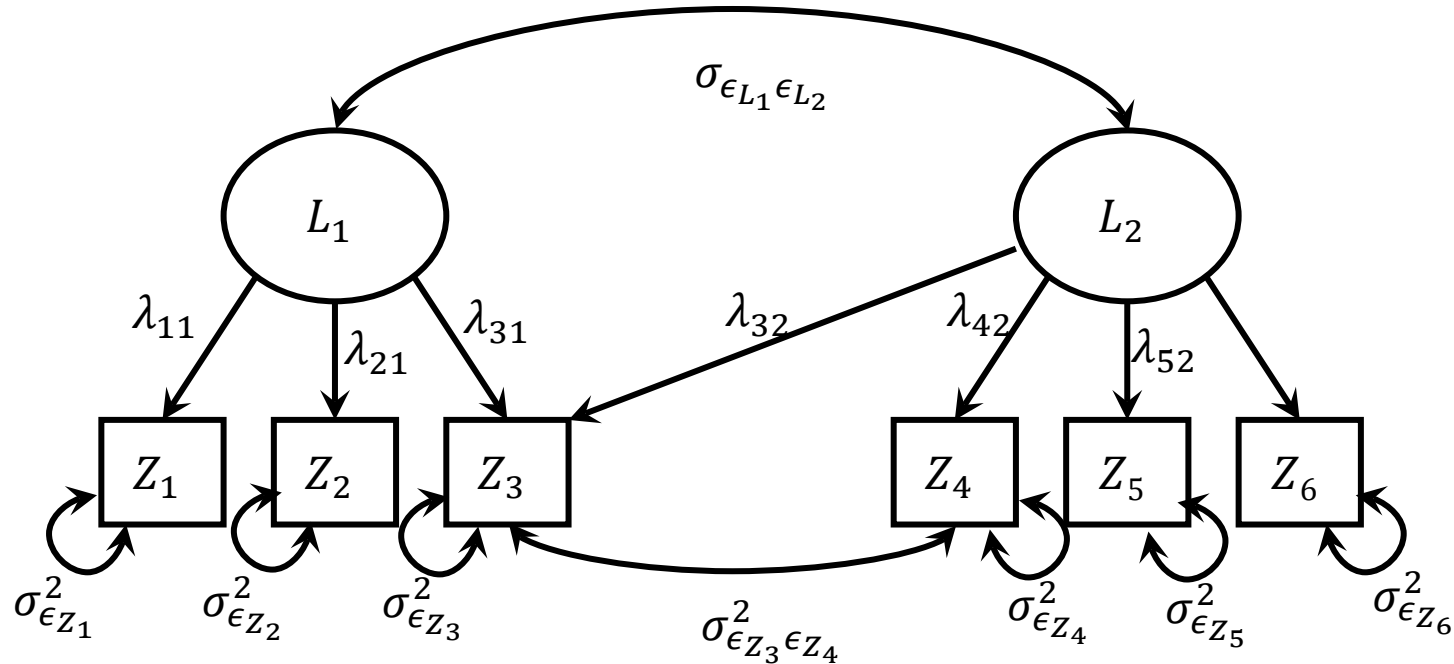
Notation differentiates latent variables and deviances, but many do not.
Deviances could be considered latent variables/factors



A shorthand you will see, but should not use

Many researchers will cut corners (and some textbooks even recommend this approach) **omitting deviances**

Do not do this, but know that diagrams like this don't actually mean what they say



Estimating Equations

The latent variables are all exogenous, so they just get an intercept and error

$$L_j = \alpha_{L_j} + \epsilon_{L_j}$$

The equations for the observed variables are more complex.

For Z s loading on a single latent variable...

$$Z_j = \alpha_{Z_j} + \lambda_{jk}L_k + \epsilon_{Z_j}$$

Note: the use of j and k here is a little vague, use path diagrams and matrices to determine

For Z s loading on a two latent variables...

$$Z_j = \alpha_{Z_j} + \lambda_{jk}L_k + \lambda_{jl}L_l + \epsilon_{Z_j}$$

In matrix form:

$$\mathbf{Z} = \boldsymbol{\alpha}_Z + \boldsymbol{\Lambda}\mathbf{L} + \boldsymbol{\epsilon}_Z$$

Matrices of a CFA

$\mathbf{L}$ is a $N_L \times 1$ vector of latent variables $\mathbf{L} = \begin{bmatrix} L_1 \\ \vdots \\ L_{N_L} \end{bmatrix}$

$\mathbf{Z}$ is a $N_Z \times 1$ vector of observed variables $\mathbf{Z} = \begin{bmatrix} Z_1 \\ \vdots \\ Z_{N_Z} \end{bmatrix}$

$\mathbf{\Lambda}$ is a $N_Z \times N_L$ matrix of factor loadings $\mathbf{\Lambda} = \begin{bmatrix} \lambda_{11} & \dots & \lambda_{1N_L} \\ \lambda_{21} & \dots & \lambda_{2N_L} \\ \vdots & \ddots & \vdots \\ \lambda_{N_Z 1} & \dots & \lambda_{N_Z N_L} \end{bmatrix}$

$\mathbf{\Sigma}_{\epsilon_L \epsilon_L}$ is a $N_L \times N_L$ matrix of covariances the latent variable errors

$$\mathbf{\Sigma}_{\epsilon_L \epsilon_L} = \begin{bmatrix} \sigma_{\epsilon_{L1}}^2 & & \\ \vdots & \ddots & \\ \sigma_{\epsilon_{L1} \epsilon_{LN_L}}^2 & \dots & \sigma_{\epsilon_{LN_L}}^2 \end{bmatrix}$$

$\mathbf{\Sigma}_{\epsilon_Z \epsilon_Z}$ is a $N_Z \times N_Z$ matrix of covariances the deviances for Z

$$\mathbf{\Sigma}_{\epsilon_Z \epsilon_Z} = \begin{bmatrix} \sigma_{\epsilon_{Z1}}^2 & & \\ \vdots & \ddots & \\ \sigma_{\epsilon_{Z1} \epsilon_{ZN_Z}}^2 & \dots & \sigma_{\epsilon_{ZN_Z}}^2 \end{bmatrix}$$

Matrices of a “typical” CFA

Consider a two-factor CFA with 3 items for each factor

Λ is structured so that only one item loads onto one factor $\Lambda = \begin{bmatrix} \lambda_{11} & 0 \\ \lambda_{21} & 0 \\ \lambda_{31} & 0 \\ 0 & \lambda_{42} \\ 0 & \lambda_{52} \\ 0 & \lambda_{62} \end{bmatrix}$

$\Sigma_{\epsilon_L \epsilon_L}$ is usually full estimated (unless you want to assume factors are uncorrelated)

$$\Sigma_{\epsilon_L \epsilon_L} = \begin{bmatrix} \sigma_{L_1}^2 & \\ \sigma_{L_1 L_2} & \sigma_{L_2}^2 \end{bmatrix}$$

$\Sigma_{\epsilon_Z \epsilon_Z}$ is a diagonal matrix, meaning the variances of the ϵ_Z s are estimated, but no covariances are estimated

$$\Sigma_{\epsilon_Z \epsilon_Z} = \begin{bmatrix} \sigma_{\epsilon_{Z_1}}^2 & 0 & 0 & 0 & 0 & 0 \\ 0 & \sigma_{\epsilon_{Z_2}}^2 & 0 & 0 & 0 & 0 \\ 0 & 0 & \sigma_{\epsilon_{Z_3}}^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sigma_{\epsilon_{Z_4}}^2 & 0 & 0 \\ 0 & 0 & 0 & 0 & \sigma_{\epsilon_{Z_5}}^2 & 0 \\ 0 & 0 & 0 & 0 & 0 & \sigma_{\epsilon_{Z_6}}^2 \end{bmatrix}$$

Specifying a “typical” CFA in Lavaan

Lavaan has a special model syntax for specifying the measurement model

```
latentvariable =~ indicator1+indicator2+indicator3
```

You get to name your latent variable whatever you want, but for the indicators you use their name from the dataframe...

```
f1 =~ x1+x2+x3
```

or

```
depression =~ x1+x2+x3
```

By default all exogenous latent variables are allowed to be correlated, so you do not need additional syntax for that.

If you want to fix the correlation to be zero, it's the same as we learned with observed variables...

```
f1 ~~0*f2
```

Implied Covariance

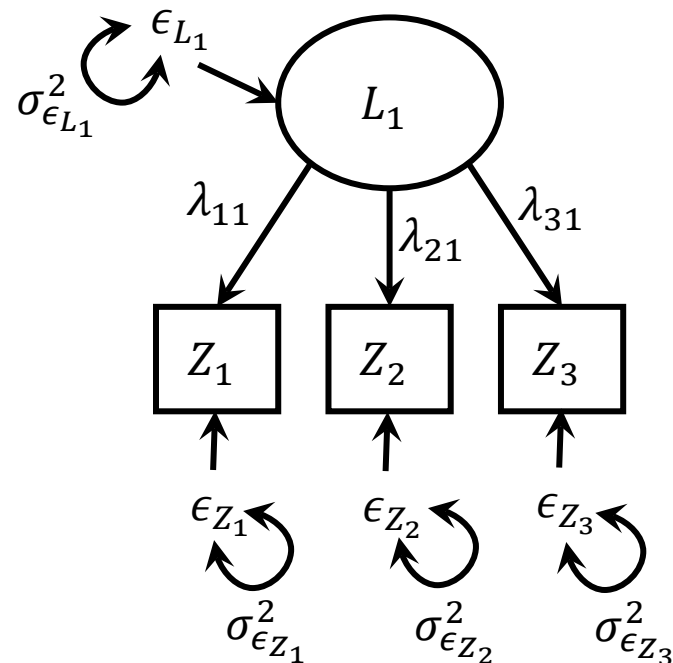
Let's consider a simple single factor model.

$$Z_1 = \alpha_{Z_1} + \lambda_{11}L_1 + \epsilon_{Z_1}$$

$$Z_2 = \alpha_{Z_2} + \lambda_{21}L_1 + \epsilon_{Z_2}$$

$$Z_3 = \alpha_{Z_3} + \lambda_{31}L_1 + \epsilon_{Z_3}$$

Work in **pairs** to find the implied covariance
between Z_1 and Z_2



Assume for simplicity that all the variances are 1

Implied Covariance

Let's consider a simple single factor model.

Covariance algebra:

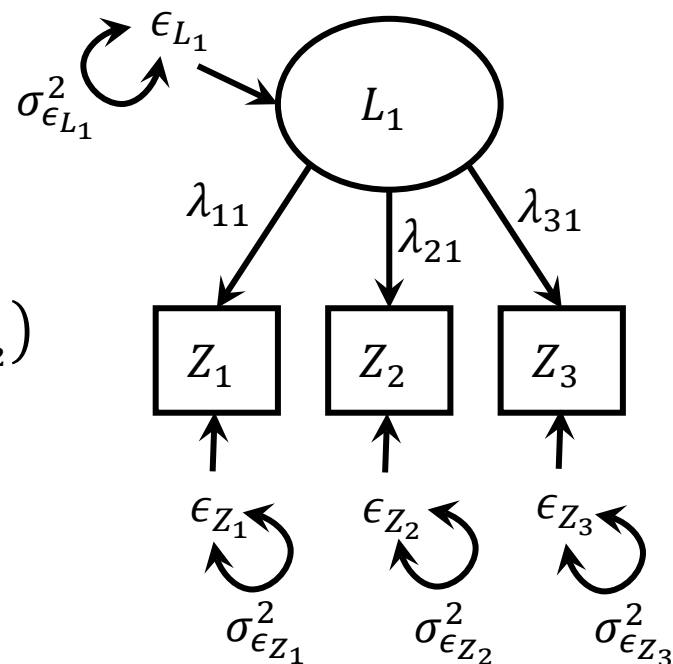
$$COV(Z_1, Z_2) = COV(\lambda_{11}L_1 + \epsilon_{Z_1}, \lambda_{21}L_1 + \epsilon_{Z_2})$$

$$= COV(\lambda_{11}L_1, \lambda_{21}L_1) + COV(\lambda_{11}L_1, \epsilon_{Z_2}) \\ + COV(\lambda_{21}L_1, \epsilon_{Z_1}) + COV(\epsilon_{Z_1}, \epsilon_{Z_2})$$

$$= \lambda_{11}\lambda_{21}COV(L_1, L_1) + \lambda_{11}COV(L_1, \epsilon_{Z_2}) \\ + \lambda_{21}COV(L_1, \epsilon_{Z_1}) + COV(\epsilon_{Z_1}, \epsilon_{Z_2})$$

$$= \lambda_1\lambda_2 1 + 0 + 0 + 0$$

Path tracing: $\lambda_1\lambda_2$



Extra: To incorporate non-zero variances in path tracing, circle through the variance on a variable when you go through $\lambda_1\lambda_2\phi_{11}$

Implied Covariance

Let's consider a simple single factor model.

$$Z_1 = \alpha_{Z_1} + \lambda_{11}L_1 + \epsilon_{Z_1}$$

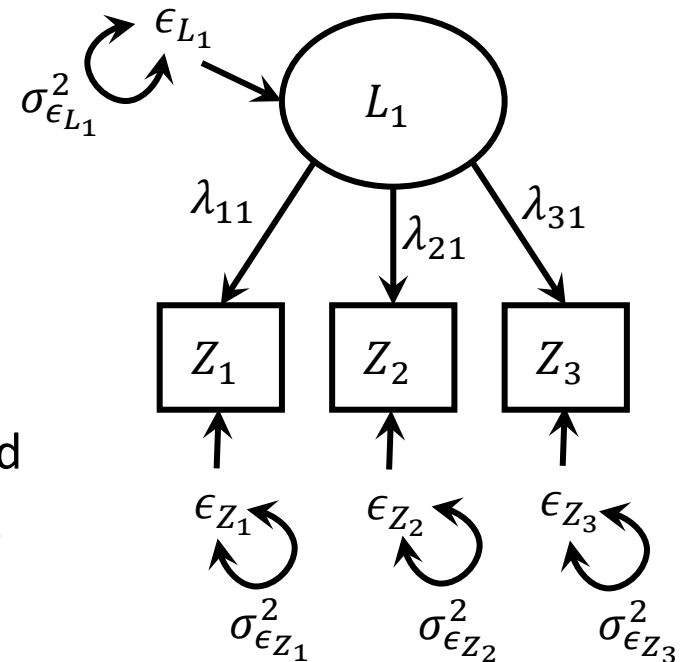
$$Z_2 = \alpha_{Z_2} + \lambda_{21}L_1 + \epsilon_{Z_2}$$

$$Z_3 = \alpha_{Z_3} + \lambda_{31}L_1 + \epsilon_{Z_3}$$

The CFA implies that the **only reason** Z_1 , Z_2 , and Z_3 are related is because they share a cause: L_1

This model does not allow for any additional sources of covariance among items

The latent variance is assumed to represent an underlying construct being measured.



Additional covariance?

What if we have reason to believe there are additional sources of covariance between items?

Consider the following items, all rated on a scale from 0 (not at all true of me) to 10 (very true of me)

- | | |
|---------------------------------------|---|
| 1. I am sad all the time | 8. I make up my mind about things easily |
| 2. Nothing will ever work out for me | 9. I have to push myself all the time to do my schoolwork |
| 3. I do everything wrong | 10. I am tired all the time |
| 4. I have fun in many things | 11. Most days I do not feel like eating |
| 5. My family is better off without me | 12. I feel alone all the time |
| 6. I hate myself | |
| 7. I feel cranky all the time | |

Two of these items are worded “positively”. This may result in a correlated residual due to **acquiescence bias** which is the tendency to use the upper part of the scale.

Correlated Deviances

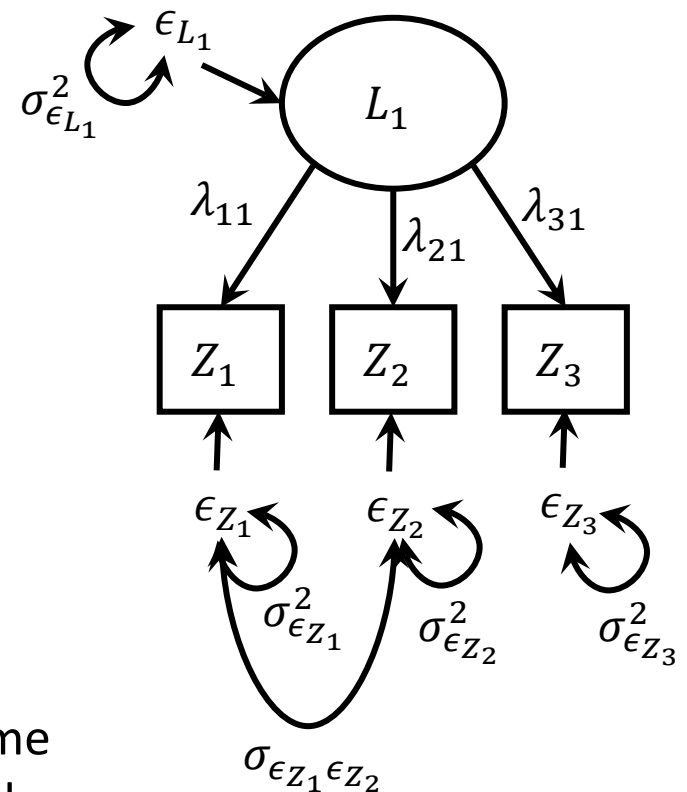
You can add a correlation among the deviances in your model to account for **local dependence**.

This does not change your estimating equations, but it does change the $\Sigma_{\epsilon_Z \epsilon_Z}$ matrix.

$$\Sigma_{\epsilon_Z \epsilon_Z} = \begin{bmatrix} \sigma_{\epsilon_{Z_1}}^2 & & \\ \sigma_{\epsilon_{Z_1} \epsilon_{Z_2}} & \sigma_{\epsilon_{Z_2}}^2 & \\ 0 & 0 & \sigma_{\epsilon_{Z_3}}^2 \end{bmatrix}$$

Adding correlated deviances in lavaan is the same as we learned with the observed variable models

`x1~~x2`



Warnings on Correlated Deviances

Too many correlated deviances can cause problems for your model and interpretation:

- Non-identification
- Low power
- Change the *meaning* of your latent variables

As we will discuss later, **modification indices** will often suggest adding correlated residuals

It's the easiest way to improve local fit

But there are other ways to approach the problem

Consider broader factors that might be needed (e.g., methods factor)

Dropping redundant items

Not all measures are “unidimensional”

Remember that latent factors are assumed to be **unidimensional**

Related to only a single dimension/construct/concept

Items are not assumed to be unidimensional

There may be multiple reasons you score high on an item

Consider this item from the CES-D: “I had trouble keeping my mind on what I was doing.”

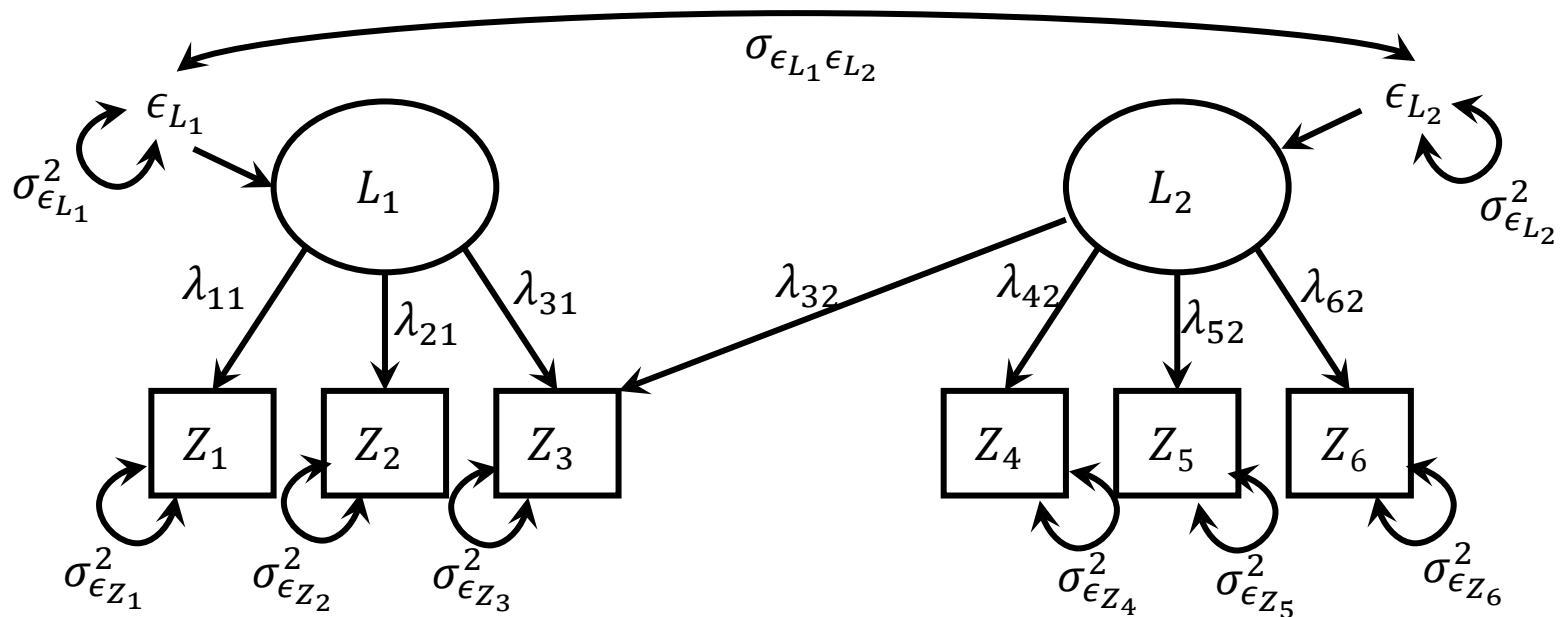
You might score high on this item because you are high on **depression** or you might score high on this item because you score high on **anxiety**

Cross-loadings

Cross loadings are a way that we can integrate multidimensionality into our models.

A cross-loading is just when we allow an item to load on multiple latent factors

Cross-loadings are useful but can cause identification problems



Specifying cross-loadings in lavaan

```
dep =~ x1+x2+x3  
anx =~ x3+x4+x5+x6
```

Cross-loading vs. Measurement Error?

It may feel hard to differentiate between cross-loadings and measurement error.

Both can lead to multiple explanations for why a person would score high on a particular item.

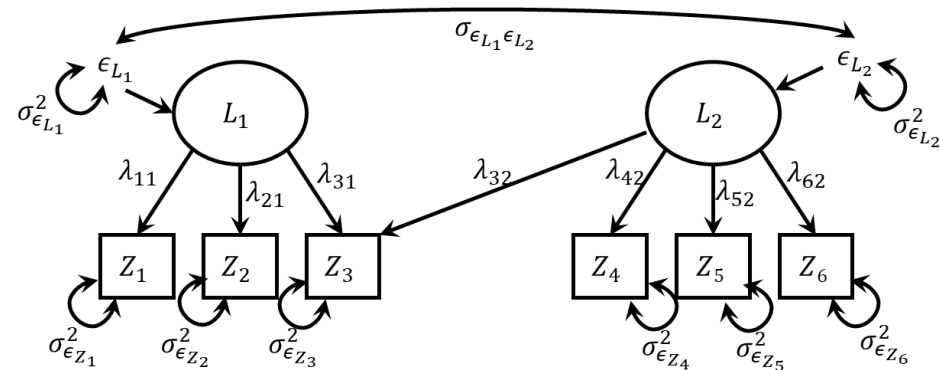
The difference is about what's going on with the **other items**

If the reason is **unique** to this item (i.e., not shared with any other item), it's measurement error

If the reason is **shared** with other items, it's a cross-loading

Notice how our example assumes there are other items that measure anxiety.

If those items don't exist, this explained variance would be incorporated into the deviance



Three Important Single Factor CFAs

There are three specific types of single factor CFAs that are worth discussing more in-depth

- **parallel models:** same factor loading, same error variances
- **tau-equivalent models:** same factor loadings, different error variances
- **congeneric models:** different factor loadings, different error variances

These models make specific restrictions on different parameters in a single-factor CFA (can be generalized to more factors)

Parallel Model

In a parallel model we assume that the factor loading is equal for all items, and the variance of the deviance is equal for all items.

Factor loading restriction:

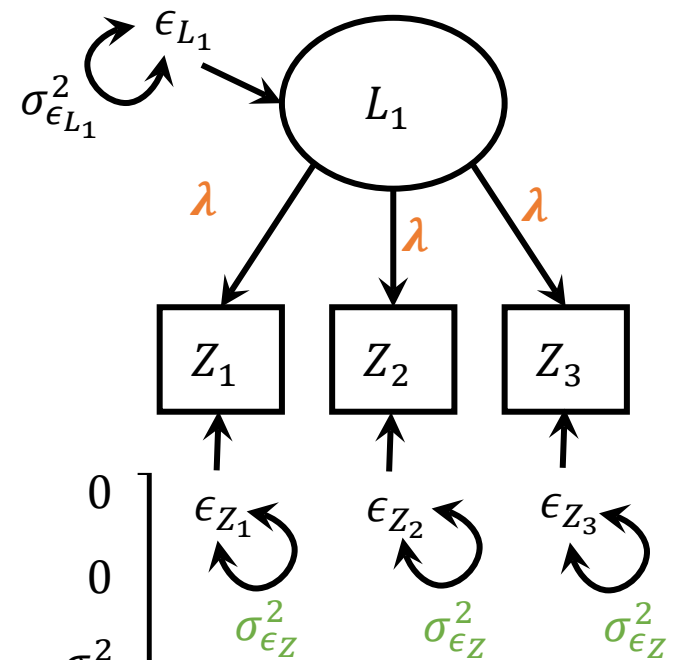
$$\lambda_{11} = \lambda_{21} = \lambda_{31} = \lambda$$

$$\Lambda = \begin{bmatrix} \lambda \\ \lambda \\ \lambda \end{bmatrix}$$

Variance restriction:

$$\sigma_{\epsilon_{Z_1}}^2 = \sigma_{\epsilon_{Z_2}}^2 = \sigma_{\epsilon_{Z_3}}^2 = \sigma_{\epsilon_Z}^2$$

$$\Sigma_{\epsilon_Z \epsilon_Z} = \begin{bmatrix} \sigma_{\epsilon_Z}^2 & 0 & 0 \\ 0 & \sigma_{\epsilon_Z}^2 & 0 \\ 0 & 0 & \sigma_{\epsilon_Z}^2 \end{bmatrix}$$



A parallel model implies that:

- Each item is an equally good indicator of the latent variable
- The amount of explained variance in each item is equal across items
- Amount of measurement error is equal across items

Specifying a Parallel Model

```
parallelsyntax <- `
  #fix factor loadings to be equal
  f1 =~lambda*x1+lambda*x2+lambda*x3
  # fix variance of deviances to be equal
  x1~~theta*x1
  x2~~theta*x2
  x3~~theta*x3
`
```

Note: Technically there is a little more you need to do, but we'll touch base on that soon

Tau-Equivalent Model

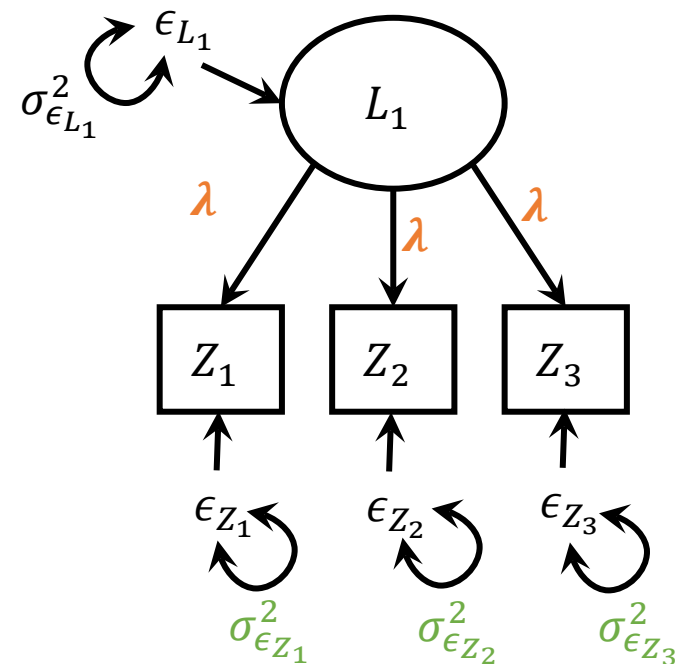
In a tau-equivalent model we assume that the factor loading is equal for all items, **but the variance of the deviance is allowed to differ.**

Factor loading restriction:

$$\lambda_{11} = \lambda_{21} = \lambda_{31} = \lambda$$

$$\Lambda = \begin{bmatrix} \lambda \\ \lambda \\ \lambda \end{bmatrix}$$

$$\Sigma_{\epsilon_Z \epsilon_Z} = \begin{bmatrix} \sigma_{\epsilon_{Z_1}}^2 & 0 & 0 \\ 0 & \sigma_{\epsilon_{Z_2}}^2 & 0 \\ 0 & 0 & \sigma_{\epsilon_{Z_3}}^2 \end{bmatrix}$$



A tau-equivalent model implies that:

- Each item is an equally good indicator of the latent variable
- The amount of explained variance in each item **may differ** across items
- Amount of measurement error is **may differ** across items

Specifying a Tau-Equivalent Model

```
taueqsyntax <- `
  #fix factor loadings to be equal
  f1 =~lambda*x1+lambda*x2+lambda*x3
`
```

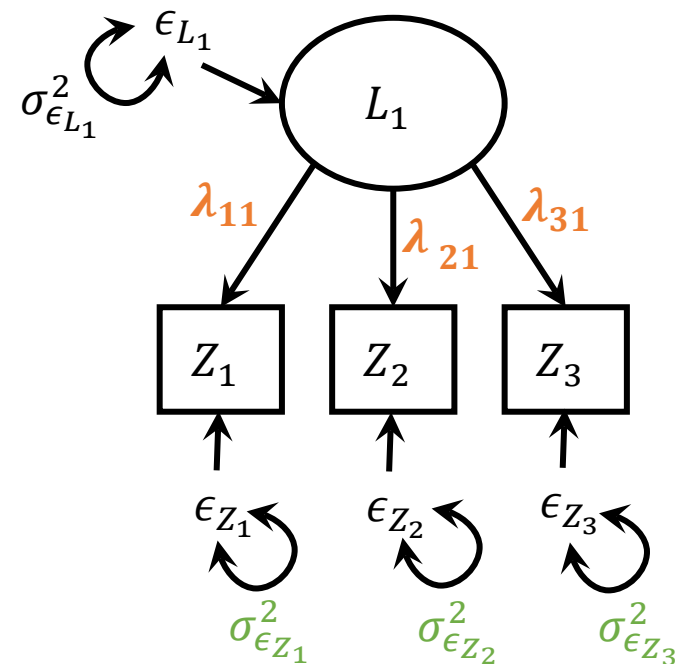
Note: Technically there is a little more you need to do, but we'll touch base on that soon

Congeneric Model

In a congeneric model we assume that the factor loading **may differ** for all items, **and the variance of the deviance is allowed to differ**.

$$\Lambda = \begin{bmatrix} \lambda_{11} \\ \lambda_{21} \\ \lambda_{31} \end{bmatrix}$$

$$\Sigma_{\epsilon_Z \epsilon_Z} = \begin{bmatrix} \sigma_{\epsilon_{Z_1}}^2 & 0 & 0 \\ 0 & \sigma_{\epsilon_{Z_2}}^2 & 0 \\ 0 & 0 & \sigma_{\epsilon_{Z_3}}^2 \end{bmatrix}$$



Congeneric models are likely the most common in psychology

Next time we will explore testing the difference between these models with data